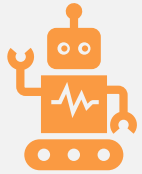


Robotics and Ethical Issues

9 Computing Tech

Learning Intention

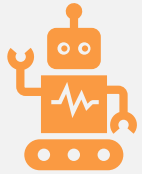


Investigate the development and impact of a mechatronic, robotic and/or automated system in a real-world application



Consider the social impacts and ethical and legal responsibilities associated with mechatronic and automated systems

Success Criteria

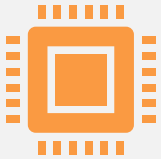


I can discuss ethical issues around the use of robotics.



I can discuss the impact of mechatronic, robotic and/or automated systems on society.

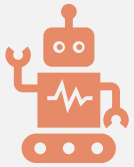
Robots vs Automated Systems



Robot: programmable machine that can perform a variety of tasks



Automated system: designed to perform one specific task



Robots are more flexible; automated systems are more specialised

What is Robotics?

Robotics: study and use of robots

Robots are automatically guided machines

Can perform tasks independently using sensors and actuators

Why Use Robots?

Perform **repetitive tasks**

Operate in **dangerous environments**

Provide **high precision and accuracy**

Can do tasks **humans cannot**

Characteristics of Robots



Programmability: can be coded to perform tasks



Mechanical capability: perform physical actions



Flexibility: can adapt or be reprogrammed

Intelligent Robots



Use **sensors** to gather data



Can **process information** and respond

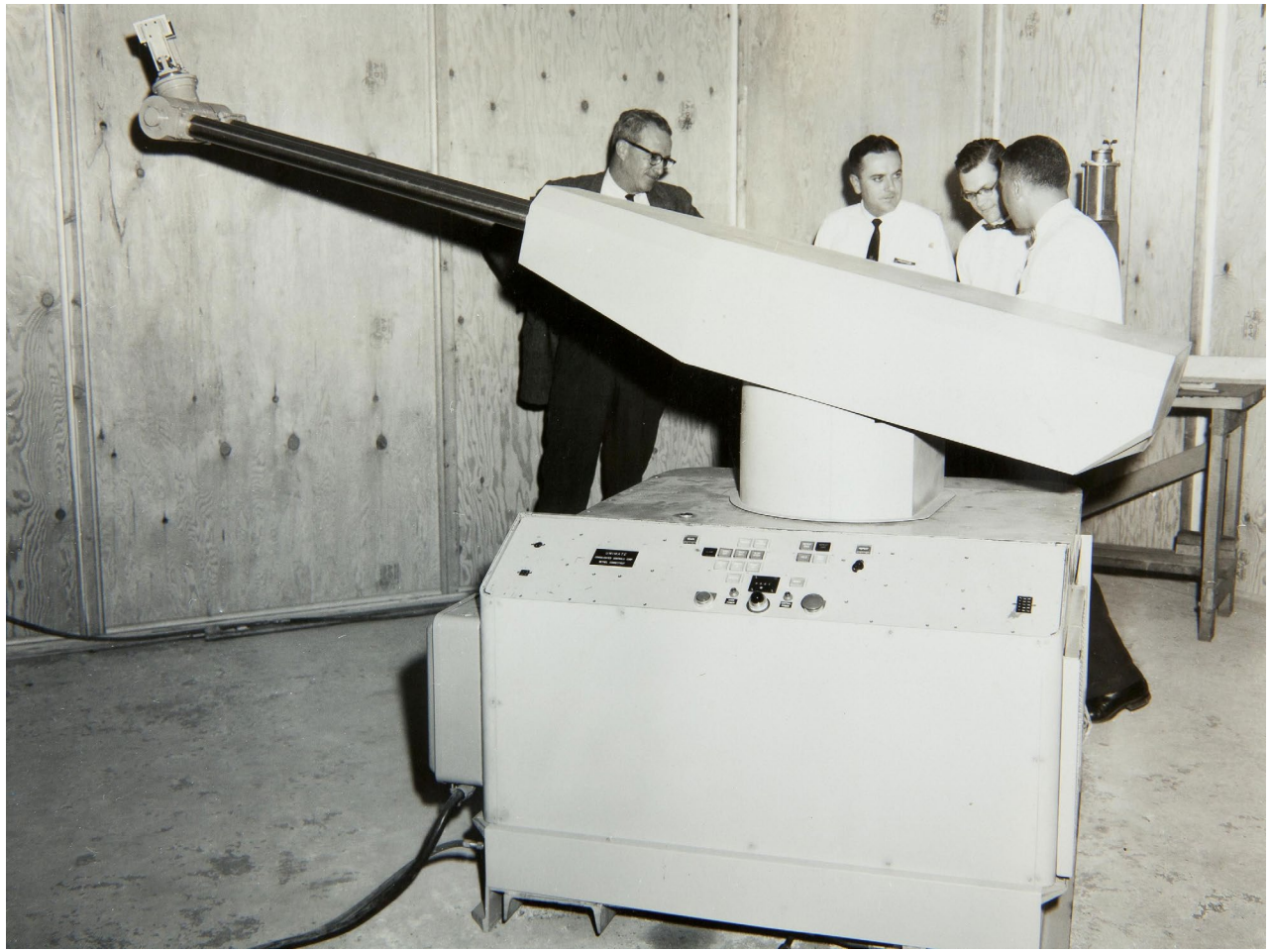


May simulate human behaviours (AI)



Examples: vision, touch, sound input

History of Robots



Early robots: **assembly line machines**

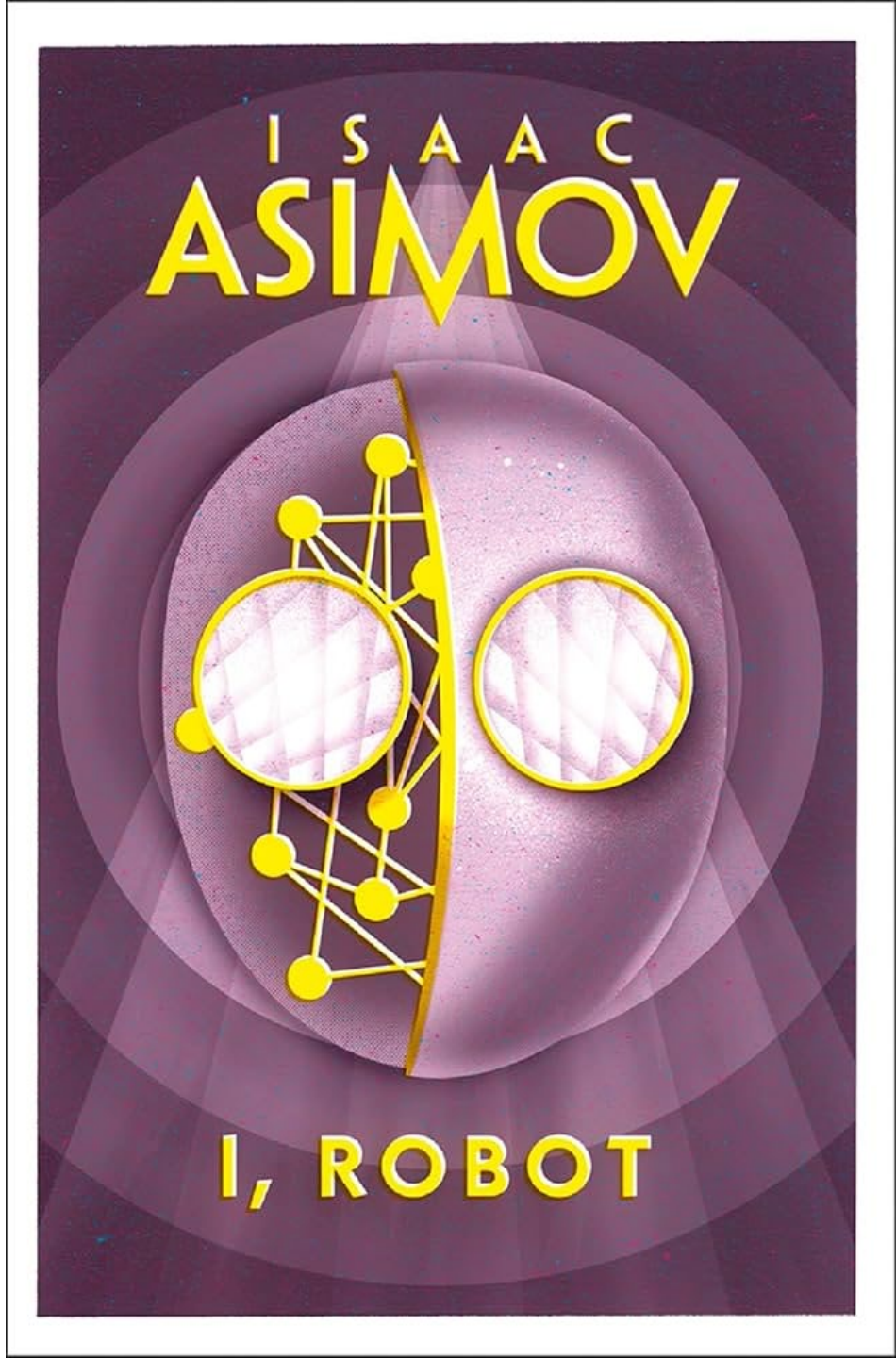
1961: first industrial robot (**UNIMATE**)

Modern robots: include sensors and AI

Inspired by earlier ideas (e.g. mechanical devices, fiction)

Asimov's Laws of Robotics

- 1. **A robot may not injure a human being** or, through inaction, allow a human being to come to harm.
- 2. **A robot must obey orders** given it by human beings except where such orders would conflict with the First Law.
- 3. **A robot must protect its own existence** as long as such protection does not conflict with the First or Second Law.



Types of Robots

Stationary robots: fixed in place (e.g. robotic arms)

Mobile robots: move through environments

Used depending on task requirements

Industrial Robots



Used in manufacturing



Tasks include:

welding
assembly
spray painting



High speed and precision

Mobile & Domestic Robots

Mobile robots: rescue, exploration, bomb disposal

Domestic robots: vacuum cleaners, security systems

Increasing use in homes and everyday life

Machines Outsmarting Humans?

Some creations challenge human capabilities:

- Robots that open doors and recharge autonomously
- Computer viruses resistant to removal
- Predator drones capable of autonomous attacks

Scientists debate limits on AI research to prevent loss of human control

Concerns: social disruption, dangerous consequences, workforce transformation

Ethical and Safety Concerns

Autonomous robots raise questions about responsibility in warfare

AI that mimics humans could be misused for crime or privacy invasion

Technological progress may displace jobs and require humans to adapt

Researchers focus on controlling AI while preventing societal harm

AI in Society

OpenAI & ChatGPT: generate human-like text and code

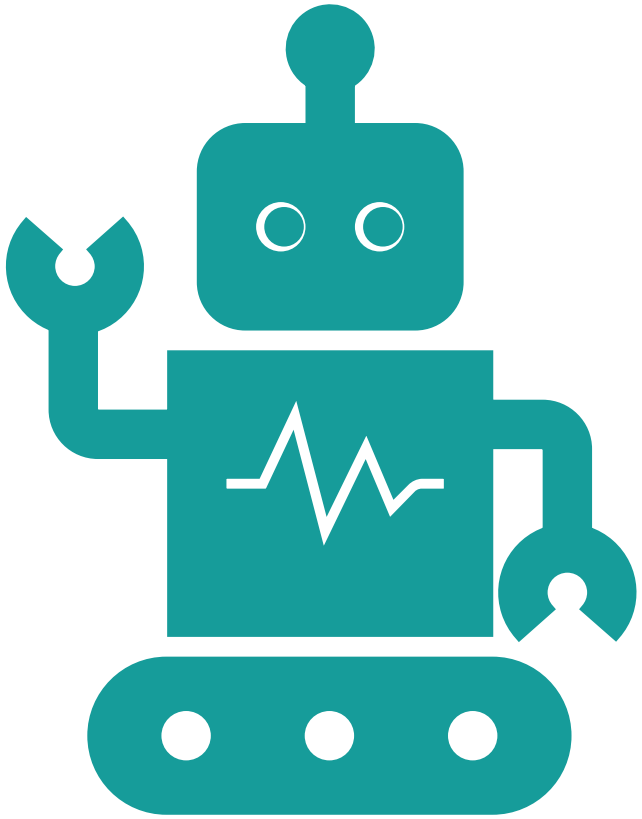
DALL·E: creates realistic images from text prompts

Concerns in education: AI can complete essays, challenging assessment integrity

Engineers are integrating AI into robots, raising new human–machine interaction questions

Class Discussion

Are companies responsible for human workers who are made obsolete by new technologies?



Class Discussion

Who is responsible if an autonomous robot causes harm?



Class Discussion

Robots should have the same rights as humans if they can think and feel.

